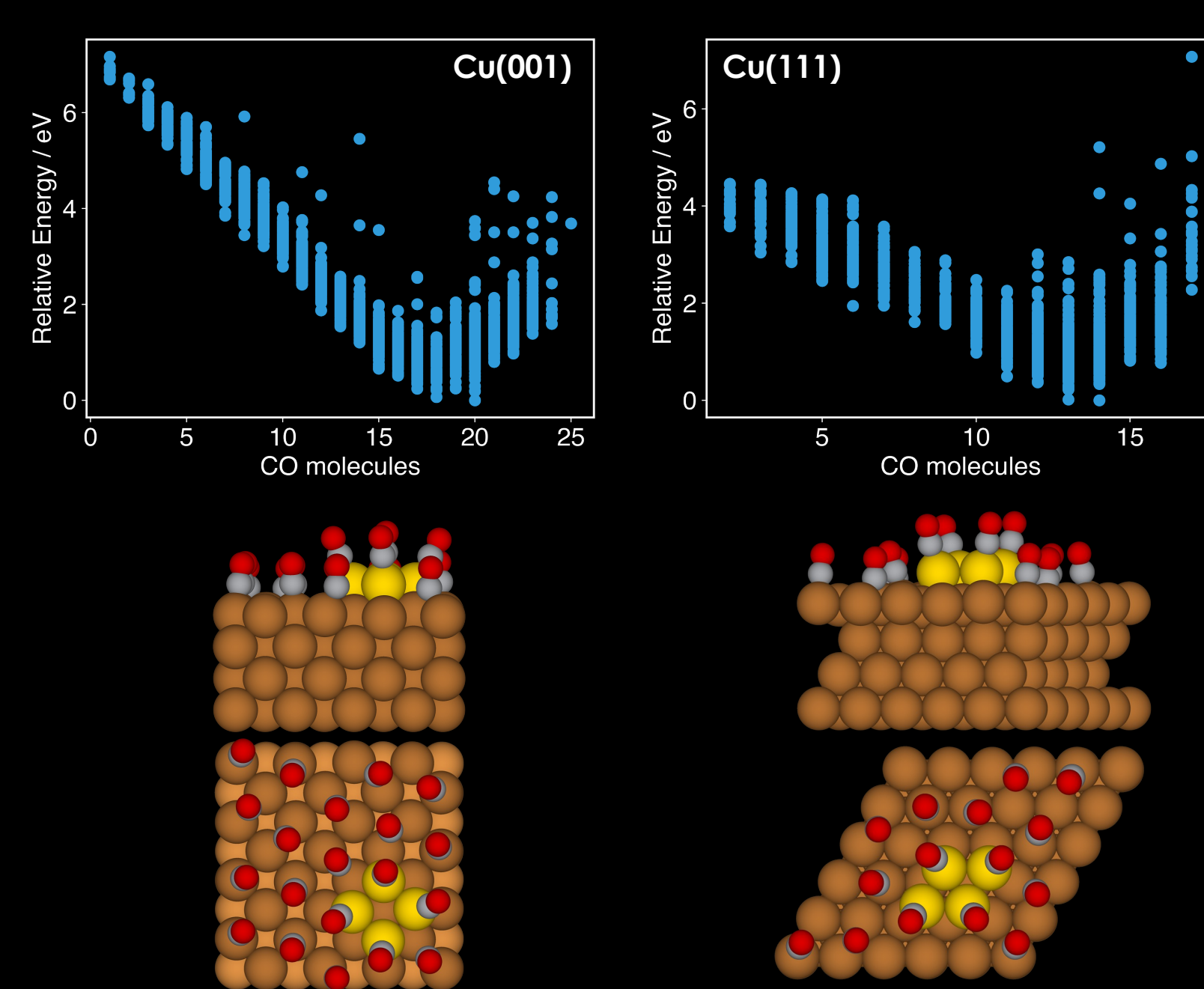
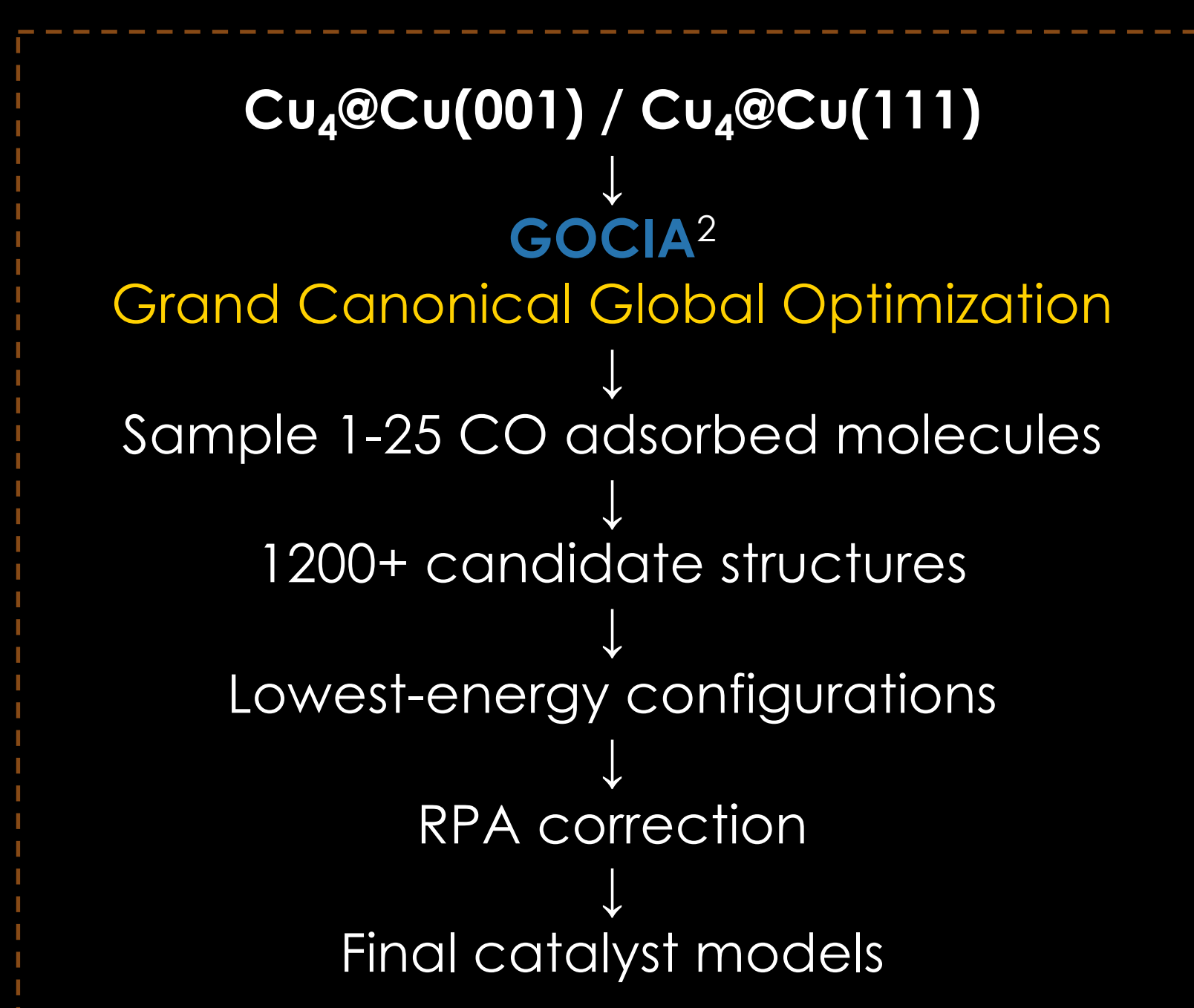


From Ideal to Realistic Cu Surfaces: Impact of Cu₄ Active Sites under CO Coverage in CO-to-Ethanol Conversion

CO COVERAGE SEARCH



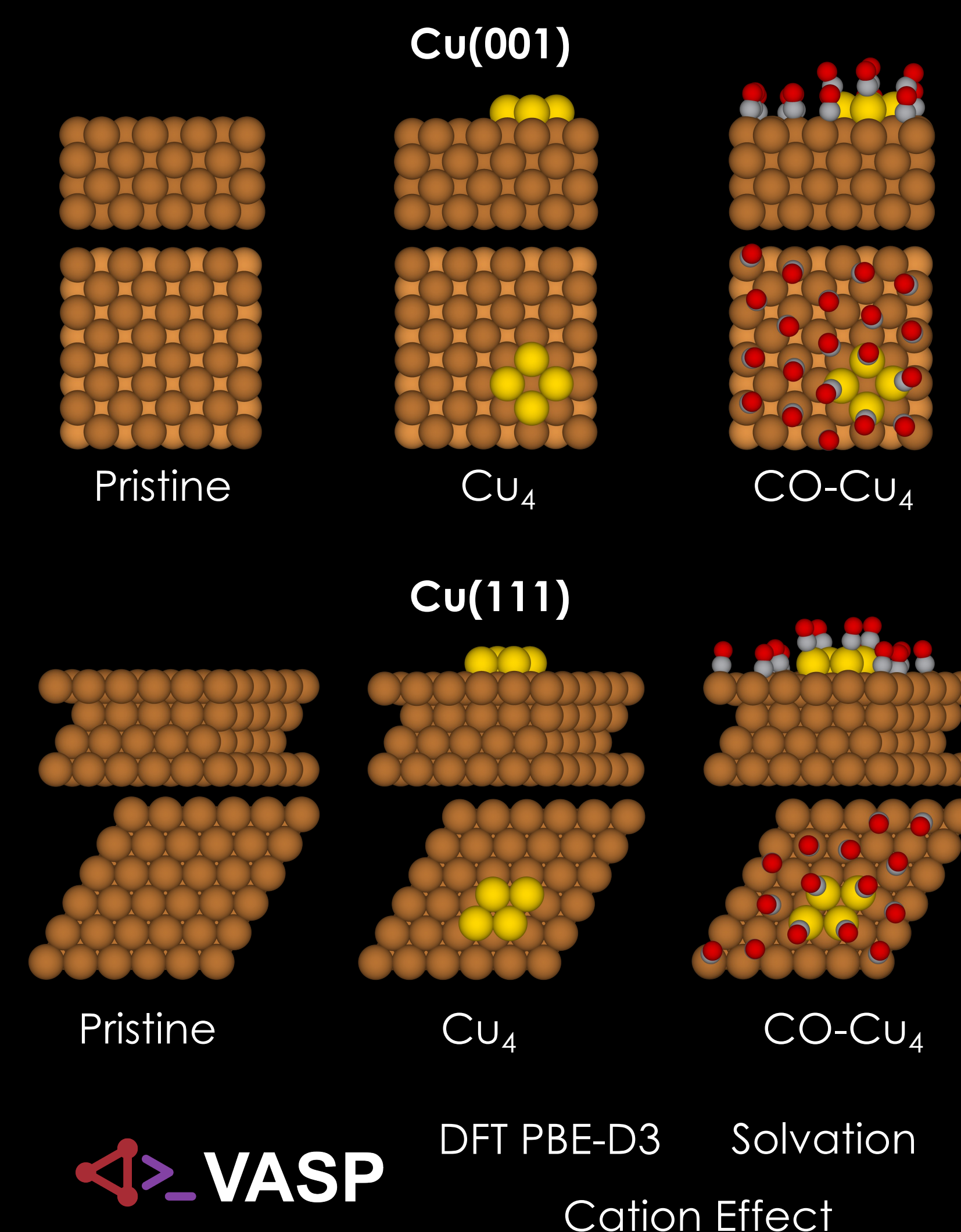
Copper (Cu) is the most effective metal catalyst for CO conversion into multicarbon products. CO is a key intermediate in CO₂ electroreduction and governs the critical C-C coupling steps leading to C₂ products. Most computational studies focus on ideal Cu surfaces, but real catalysts operate under reaction conditions where surface restructuring, adsorbates, and undercoordinated sites are present. Among these active sites, Cu₄ nanoislands have been proposed as highly active centers for ethanol production¹ but...

Does the Cu₄ nanoisland remain catalytically active under realistic CO coverages?



SURFACE MODELS

To determine the effects of both Cu₄ nanoislands and CO coverage on ethanol formation, three representative models were considered for both Cu(001) and Cu(111):



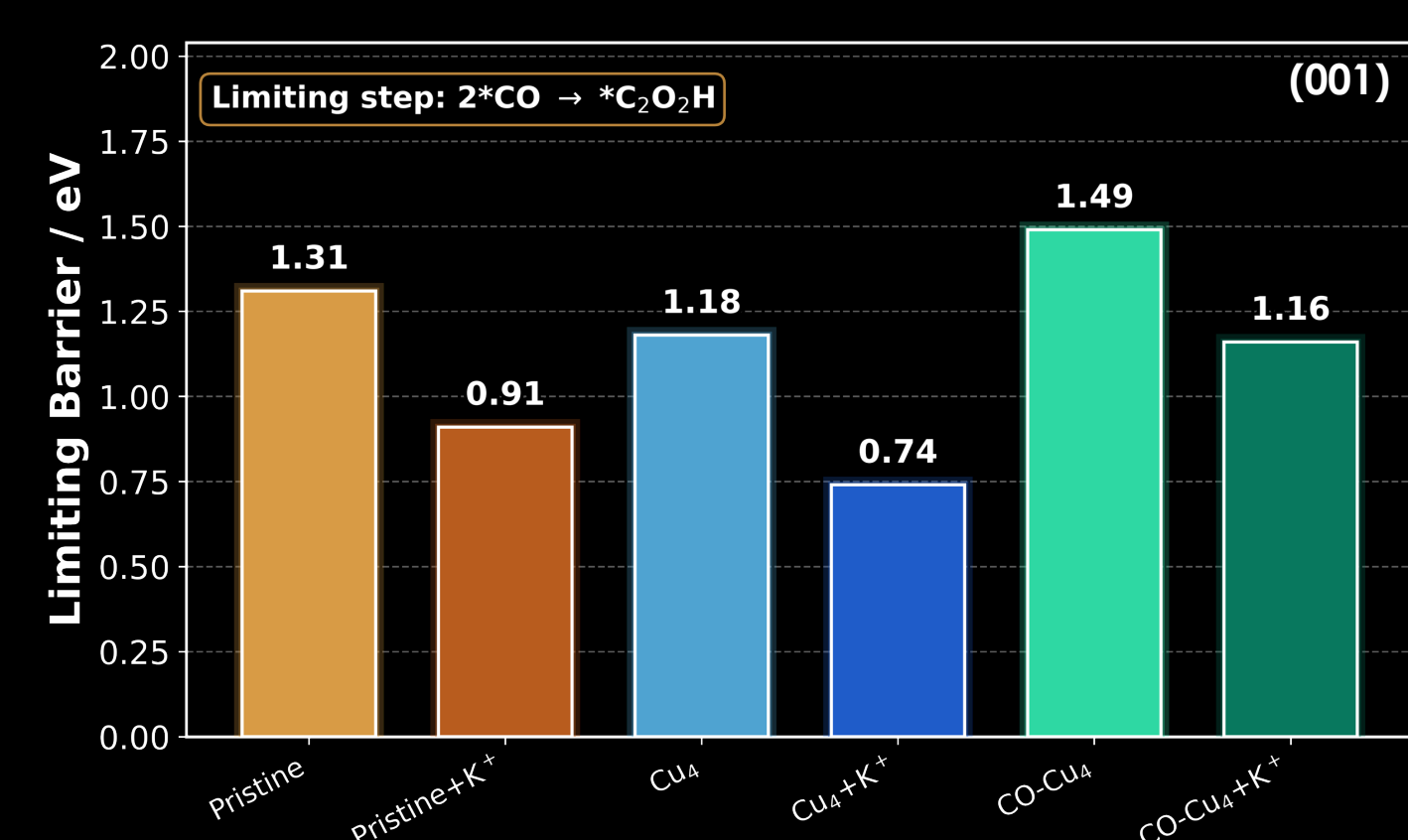
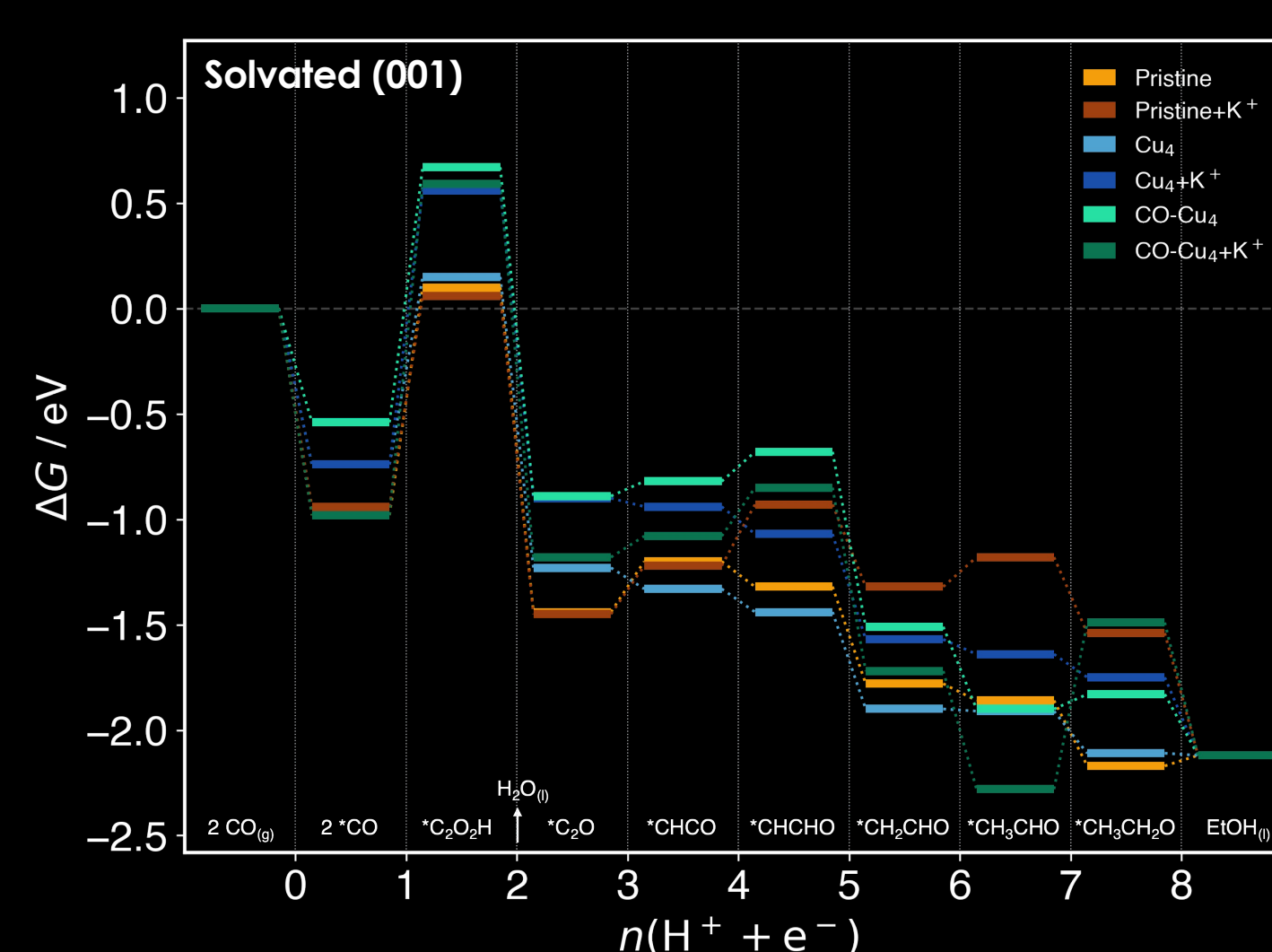
RESULTS (001)

The C-C coupling step (2*CO → C₂O₂H) is the limiting step for all models

K⁺ strongly promotes C-C coupling, lowering the barrier by up to 0.4 eV

Cu₄ active sites stabilize the C₂O₂H intermediate, lowering the barrier

Realistic CO-covered Cu₄ sites remain catalytically active despite higher barriers

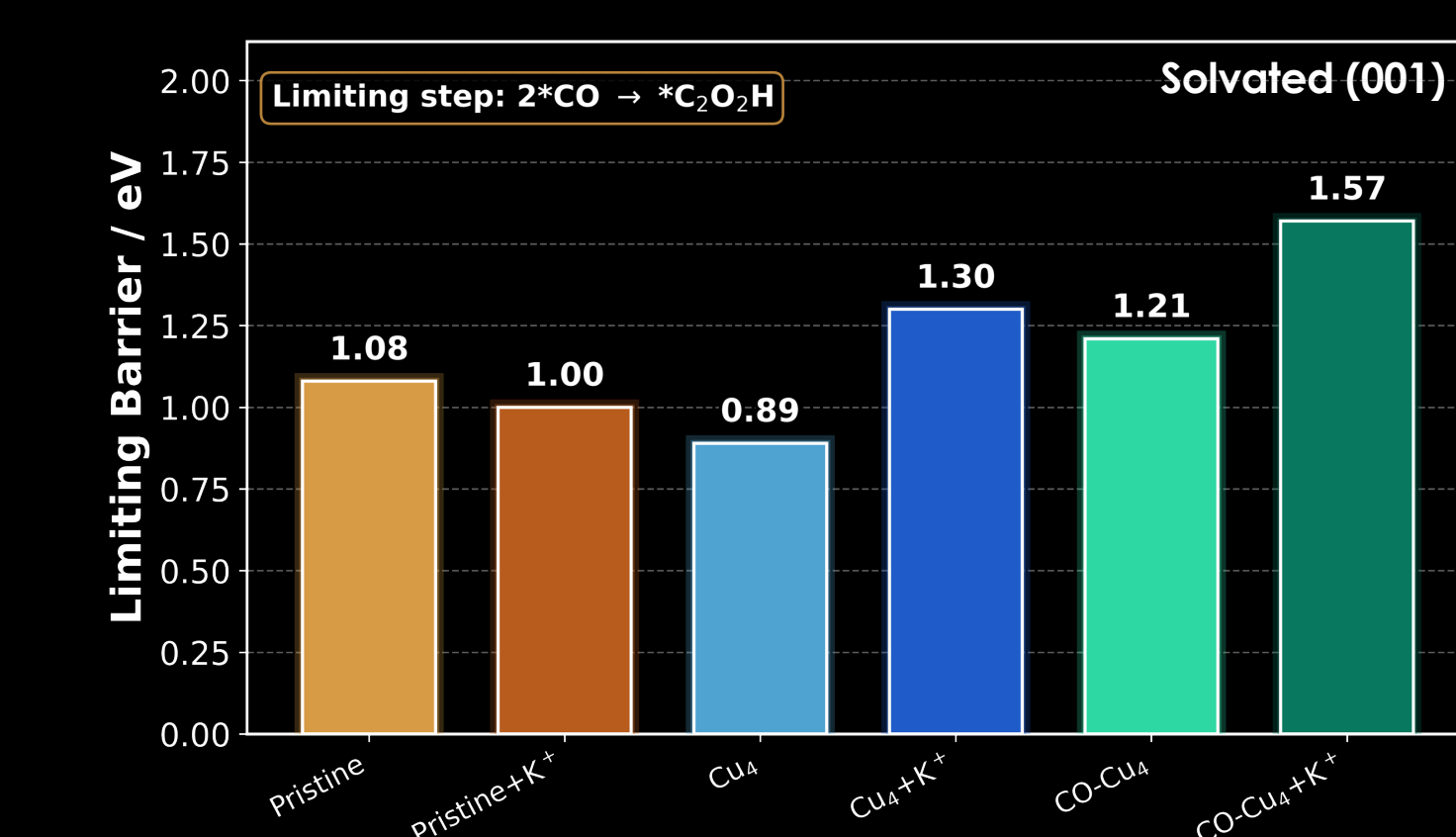


Solvation preserves C-C coupling as the rate-determining descriptor

The beneficial effect of K⁺ observed in vacuum is strongly reduced in solution

Cu₄ nanoislands reduce the limiting barrier by ca. 0.2 eV compared to pristine Cu(001)

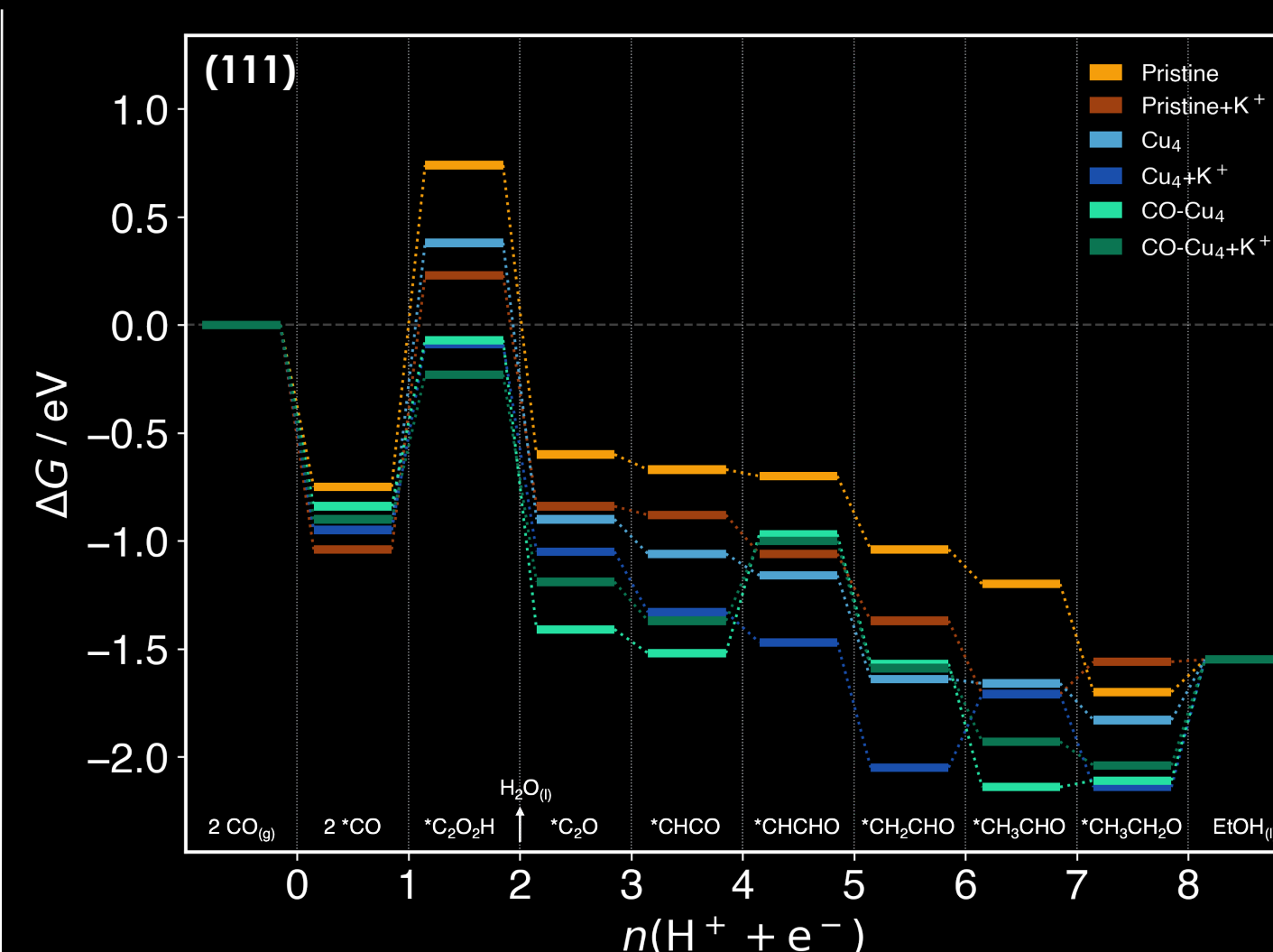
The effect of K⁺ is strongly site-dependent, becoming detrimental on Cu₄ active sites



CONCLUSIONS

CO-covered Cu₄ nanoislands remain active under realistic reaction conditions, although their catalytic behavior strongly depends on the surface orientation. On Cu(001), CO coverage generally increases the limiting barrier, whereas on Cu(111) it lowers the barrier and shifts the limiting step from C-C coupling to CHCO hydrogenation. This mechanistic change is accompanied by a spontaneous restructuring of the Cu₄ island, emphasizing the dynamic nature of realistic catalysts.

RESULTS (111)

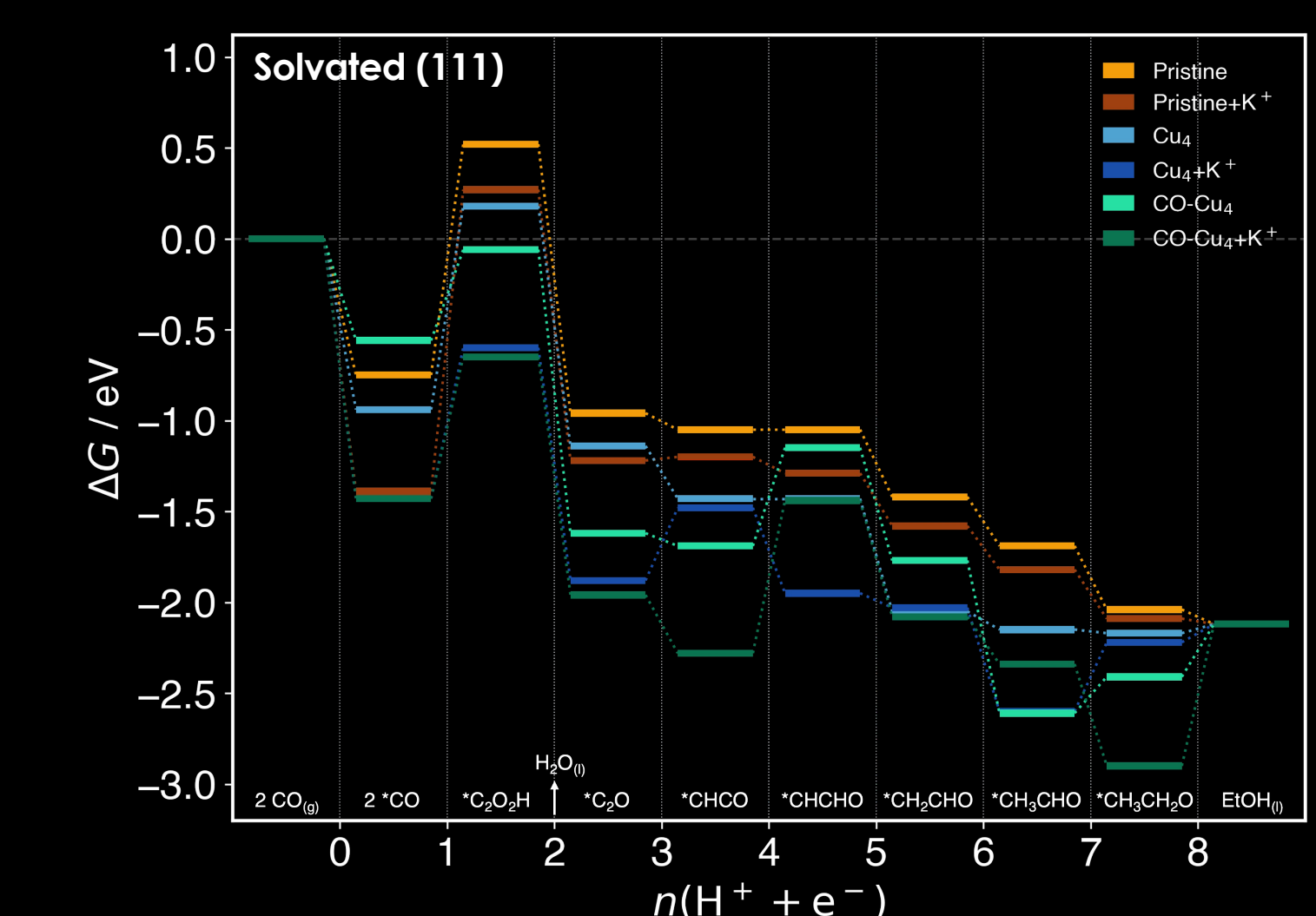
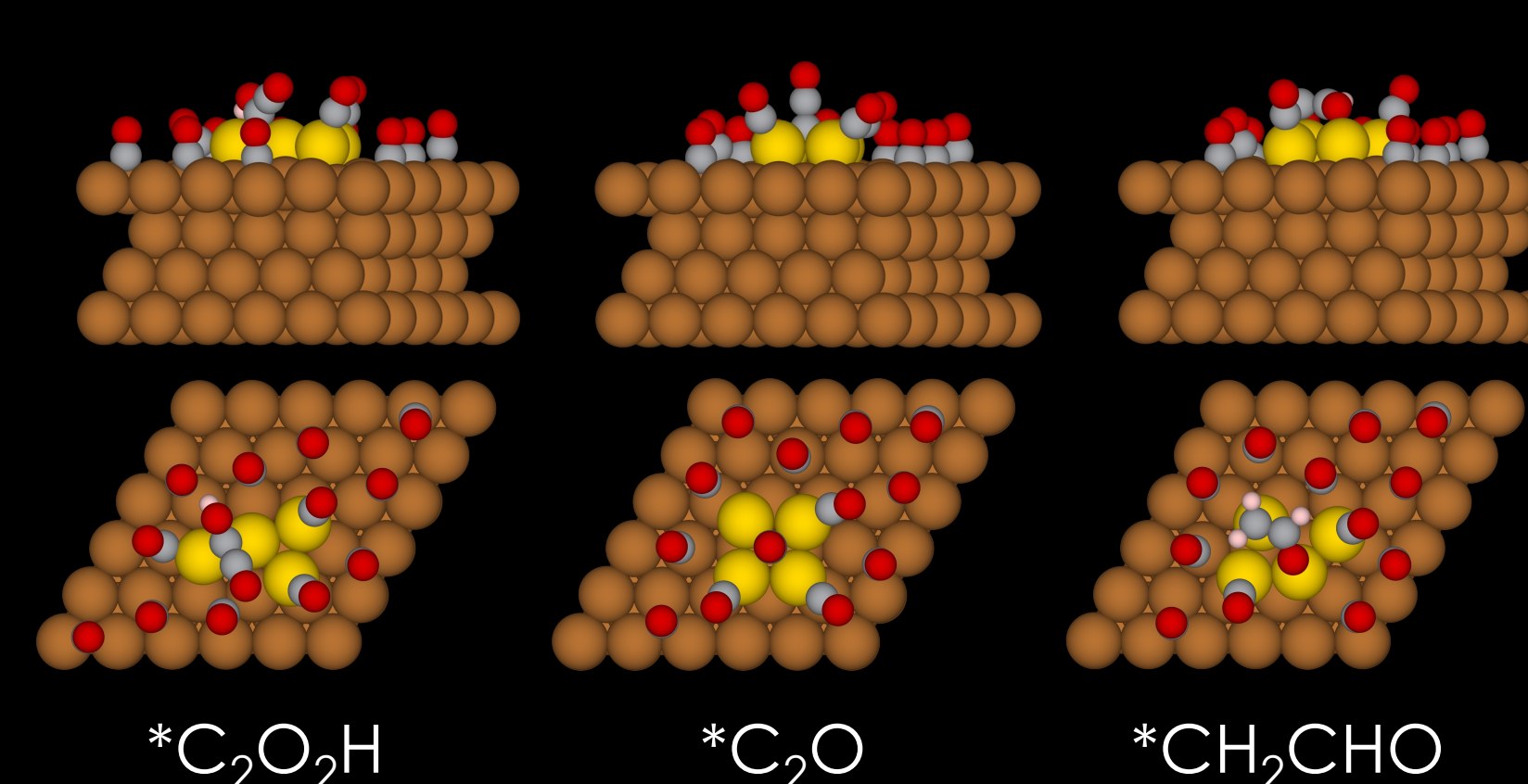


The C-C coupling step (2*CO → C₂O₂H) remains the determining step

Cu₄ nanoislands still lower the limiting barrier relative to pristine Cu(111)

CO-covered Cu₄ active sites exhibit the lowest limiting step barriers

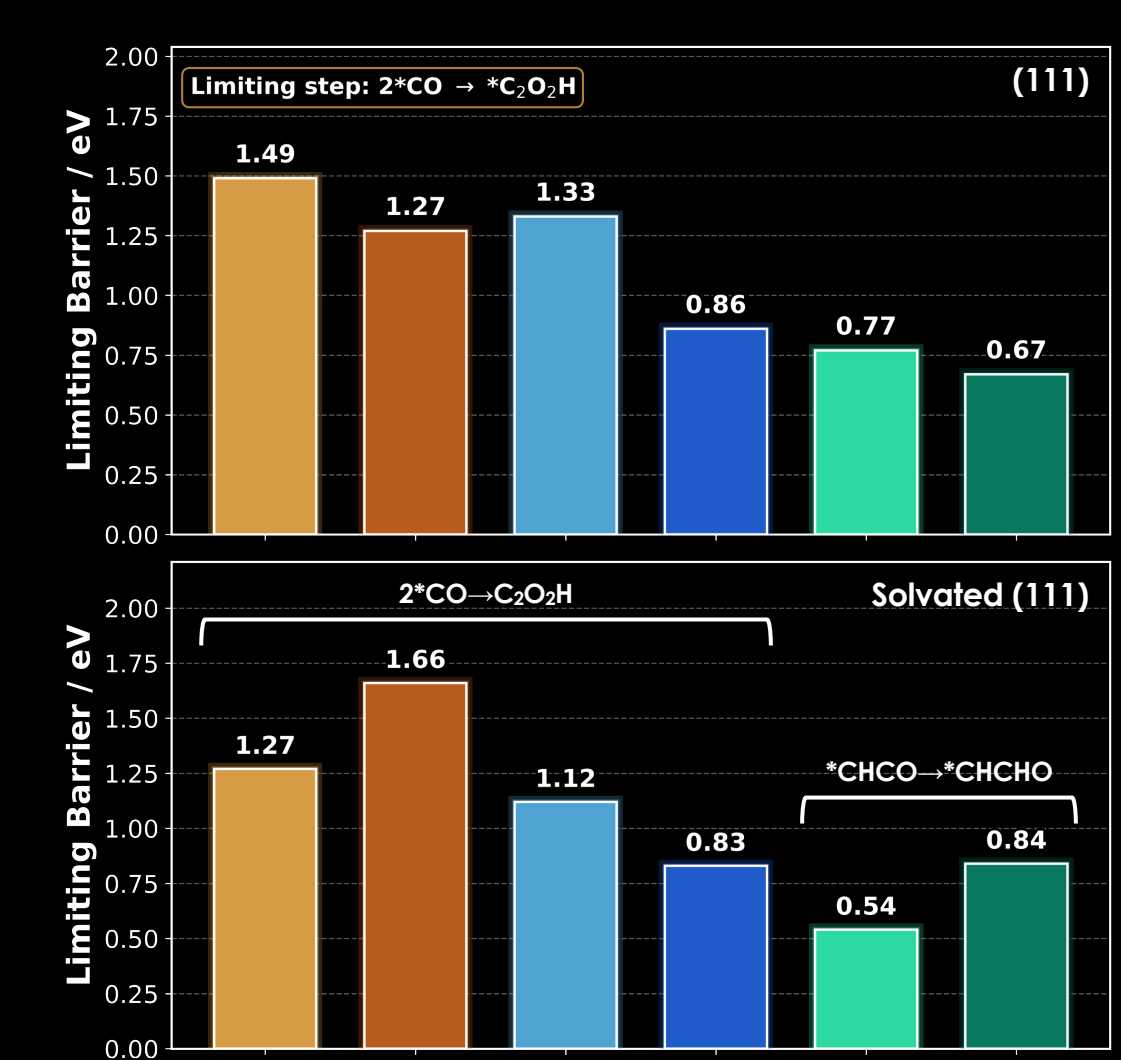
The CO-covered (111) catalyst is accompanied by a spontaneous restructuring of the Cu₄ cluster during intermediate adsorption



C-C coupling is the limiting step on pristine and idealized Cu₄ surfaces

CO-covered Cu₄ sites shift the limiting step to *CHCO → *CHCHO

Surface restructuring modifies the reaction path under realistic conditions



REFERENCES

- O. Piqué, F. Viñes, F. Illas, F. Calle-Vallejo, *ACS Catal.* **2020**, *10*, 10488-10494.
- Z. Zhang, W. Gee, R. H. Lavroff, A. N. Alexandrova, *Phys. Chem. Chem. Phys.* **2025**, *27*, 696.

